

Standard Treatment Workflow (STW)

NEONATAL HYPOGLYCEMIA

ICD-11- KB60.4

WHOM TO SCREEN FOR HYPOGLYCEMIA

- Preterm infants (<37 weeks’ gestation)
- Low-birth-weight infants (<2500 g)
- Small-for-gestational-age (SGA) infants: birth weight <10th percentile as per INTERGROWTH-21st standards
- Large-for-gestational-age (LGA) infants: birth weight >90th percentile as per INTERGROWTH-21st standards
- Infants of diabetic mothers (IDM)
- Neonates born to mothers receiving beta-blockers
- Sick neonates, including those with sepsis, shock, birth asphyxia, respiratory distress, polycythaemia, on IV fluids
- Neonates following exchange transfusion

Routine blood glucose monitoring is not required in healthy term AGA neonates

SCHEDULE OF BLOOD GLUCOSE MONITORING	
CATEGORY	TIME SCHEDULE
At-risk infants	1, 2, 6, 12, 24, 48, 72 hours
	*in IDM, monitoring can be stopped between 24-36 hours provided feeding is established
Infants on IV fluids	Every 6-8 hours
BG should be measured pre-feeding	

HOW TO MONITOR BLOOD GLUCOSE (BG)

- Measure BG using a point-of-care glucometer
- Low value (<45 mg/dL): send a blood sample to the lab
- DO NOT delay treatment for lab confirmation

BLOOD GLUCOSE <45 mg/dL

LOOK FOR THE FOLLOWING SYMPTOMS AND SIGNS

- Stupor, lethargy, limpness
- Jitteriness, tremors, convulsions
- Episodes of cyanosis, apnoea or tachypnoea
- Weak or high-pitched cry
- Unable to feed

ASYMPTOMATIC & BG ≥25 mg/dL

- Immediate supervised feeding
- Breastfeeding or a measured volume of expressed breastmilk (Formula milk if EBM not available) by paladai or gavage

RE-CHECK BG AFTER 1 HOUR

If BG ≥ 45 mg/dL

- Continue feeds
- Continue BG monitoring 6 hourly for 24 hours

Start IV glucose infusion if:

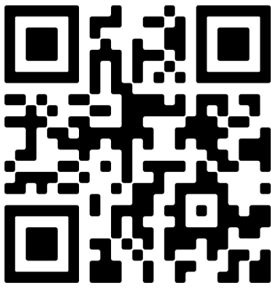
- BG < 45 mg/dL OR
- Symptoms develop

PREVENTION OF HYPOGLYCEMIA

- Support mother for early initiation and regular breastfeeding
- Maintain normothermia
- Do not feed dextrose solutions as a substitute for breastmilk

PRACTICAL POINTS

- Avoid > 12.5-15% dextrose infusion through a peripheral vein
- Maintain patency of IV cannula
- Use a syringe/infusion pump to deliver glucose
- Avoid frequent dextrose boluses
- Always search for an underlying cause – polycythemia, sepsis, meningitis, hypothermia, IUGR



GIR calculations



Heel Prick for BG estimation

SYMPTOMATIC OR BG < 25 mg/dL

- IV bolus: 2 ml/kg of 10% of dextrose slowly over 1 minute
- Start IV infusion of dextrose at a glucose infusion rate (GIR) of 6 mg/kg/min

Re-check BG every 30 min until 2 consecutive values are ≥45 mg/dL, then every 6 h

BG < 45 mg/dL

Increase GIR by 2 mg/ kg/min (maximum 12 mg/kg/min)

Persistent (GIR requirement >3-7 days) or Refractory (GIR > 10-12 mg/kg/min for 24 hours) hypoglycemia

Consider ENDOCRINE/ METABOLIC disorders & REFER to higher centre

DRUGS FOR REFRACTORY HYPOGLYCEMIA

- Hydrocortisone: 5 mg/kg/day IV in two divided doses
- Additional drugs should be administered at a tertiary care hospital with workup for underlying causes

BG ≥ 45 mg/dL

Euglycemic for 24 hours on IV fluids

- Reduce GIR by 2 mg/ kg/min every 6 hours
- Increase oral feeds
- Monitor BG every 6 hours

Stop IV fluids when euglycemic on GIR 4 mg/ kg/min and tolerating adequate enteral feeds

DO’s

- Monitor BG in all; as per protocol
- Support early and frequent breastfeeding
- Treat symptomatic hypoglycemia immediately

DON’Ts

- Do NOT substitute dextrose solutions for breastmilk
- Do NOT separate a stable mother-infant dyad solely for glucose monitoring
- Do NOT give antibiotics for hypoglycemia unless sepsis is suspected

KEY PERFORMANCE INDICATORS (KPIs)

Blood glucose screening coverage = $\frac{\text{At-risk neonates monitored for BG as per protocol} \times 100}{\text{Total at-risk neonates eligible for BG monitoring}}$ (Target ≥90%)

Timely BG re-check after treatment = $\frac{\text{Hypoglycemic episodes with repeat BG within prescribed interval} \times 100}{\text{No. of hypoglycemic episodes treated}}$ (Target ≥90%)

NEONATES WITH SYMPTOMATIC, SEVERE, RECURRENT OR PERSISTENT HYPOGLYCEMIA SHOULD RECEIVE STRUCTURED NEURODEVELOPMENTAL FOLLOW-UP

ABBREVIATIONS

AGA: Appropriate for Gestational Age
BG: Blood Glucose

EBM: Expressed Breastmilk
GIR: Glucose Infusion Rate

IDM: Infant of Diabetic Mother
IV: Intravenous

IUGR: Intrauterine Growth Restriction

REFERENCES

1. Committee on Fetus and Newborn, Adamkin DH. Postnatal glucose homeostasis in late-preterm and term infants. Pediatrics 2011; 127:575.
2. Cornblath M, Hawdon JM, Williams AF, et al. Controversies regarding definition of neonatal hypoglycemia: Suggested operational thresholds. Pediatrics 2000;105(5):1141–5.
3. Management of Neonatal Hypoglycemia. 25 November 2023. New Delhi: National Neonatology Forum of India; 2023 (NNF/2023/CPG/ 2023.5.0) Accessed from: <https://app.magicapp.org/#/guideline/6893>